Introduction

The Brier score for survival is a prediction-loss metric that combines calibration and discrimination into one number, evaluated at a chosen time horizon t. It generalises the binary Brier score to right-censored data via inverse-probability-of-censoring weights, so a model that predicts a survival probability $\hat{S}_i(t)$ for each subject can be assessed against the actual observed status at t. Unlike the C-index, which captures only ranking, the Brier score penalises miscalibrated probabilities — a model that ranks well but consistently overpredicts survival will score worse on Brier than on concordance.

Prerequisites

A working understanding of survival predictions, calibration vs discrimination, censoring, and the binary Brier score $(\hat{p} - y)^2$.

Theory

At time t, the censoring-weighted Brier score is

$$\mathrm{BS}(t) = \frac{1}{n} \sum_{i=1}^{n} w_i(t) \big(Y_i(t) - \hat{S}_i(t)\big)^2,$$

where $Y_i(t) = \mathbf{1}\{T_i > t\}$ is the observed survival indicator (when defined) and $w_i(t)$ is an inverse-probability-of-censoring weight that adjusts for the missing $Y_i(t)$ when subject i is censored before t. The integrated Brier score (IBS, also called the continuous ranked probability score for survival) averages $\mathrm{BS}(t)$ over a follow-up window and gives a single-number summary suited to model selection.

A useful reference is the Brier score of the marginal Kaplan-Meier estimator (no covariates); a model that delivers a lower IBS than the null KM has produced informative predictions.

Assumptions

The censoring weights $w_i(t)$ are correctly estimated, typically via the Kaplan-Meier of the censoring distribution, which assumes non-informative censoring conditional on covariates.

R Implementation

```r
library(pec); library(survival)

data(lung)
fit <- coxph(Surv(time, status) ~ age + sex, data = lung, x = TRUE, y = TRUE)
pec_out <- pec(list(fit), data = lung,
               formula = Surv(time, status) ~ 1,
               times = seq(100, 800, by = 100))

plot(pec_out)
crps(pec_out)
```

Output & Results

`pec()` returns time-dependent Brier scores at the requested horizons together with the null KM benchmark. `crps()` returns the integrated Brier score (continuous ranked probability score) over the requested window. The plot overlays the model and reference Brier curves; a model curve below the KM curve at every t represents predictive improvement throughout follow-up.

Interpretation

A reporting sentence: "The Cox model achieved an integrated Brier score of 0.16 over 100–800 days, compared to 0.21 for the null Kaplan-Meier; predictive improvement was largest between 200 and 500 days, where the model's time-specific Brier was about 30 % below the reference." Pair the IBS with a C-index; together they cover both calibration and discrimination.

Practical Tips

- Compare the model to the null KM at every time point you report; without that reference, an absolute Brier score is hard to interpret.
- Time-dependent Brier curves reveal when the model is most and least informative; this often identifies follow-up windows where extra covariates would help.
- For honest assessment, use cross-validation: `pec(..., splitMethod = "cv10")` performs 10-fold CV and adjusts for optimism.
- Integrated Brier (IBS) is a single-number summary suited to model ranking; report it alongside per-time-point Brier curves for transparency.
- Complement with calibration plots (predicted vs observed survival probabilities at t) and the C-index; the three together give a full predictive-performance picture.
- Brier-score weights depend on a censoring model; if censoring depends on covariates, fit a Cox model for the censoring time and use it instead of the marginal KM.

R Packages Used

`pec` for `pec()`, `crps()`, and prediction-error curves; `riskRegression` for unified discrimination, calibration, and Brier metrics with bootstrap CIs; `survival` for the Cox fit; `timeROC` for time-dependent AUCs that complement the Brier-score view.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation